

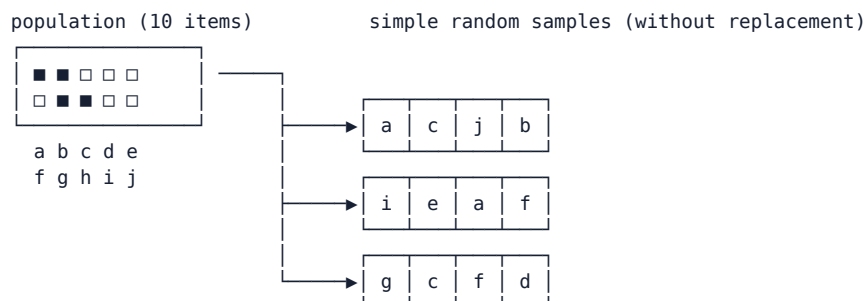
# 08 SAMPLING AND THE CENTRAL LIMIT THEOREM (lms-standard-error-clt, 19 pages, lectures 17-18)

## 1. Population vs sample (p001-p002)

| POPULATION (N)                                      | SAMPLE (n)                                         |
|-----------------------------------------------------|----------------------------------------------------|
| the ENTIRE group you want to draw conclusions about | a smaller subset, collected to observe and analyse |
| described by PARAMETERS<br>$\mu, \sigma$            | described by STATISTICS<br>$\bar{X}, s$            |
| often too large / costly / impossible to measure    | must be representative and randomly selected       |
| e.g. all 500,000 bulbs                              | e.g. 100 randomly chosen                           |

PARAMETER vs STATISTIC - the words the exam checks  
 PARAMETER = a number about the POPULATION ( $\mu, \sigma, p$ )  
 STATISTIC = a number computed FROM THE SAMPLE ( $\bar{X}, s, p^{\wedge}$ )  
 The parameter is UNKNOWN (that is why we sample).  
 The statistic is KNOWN (we compute it).

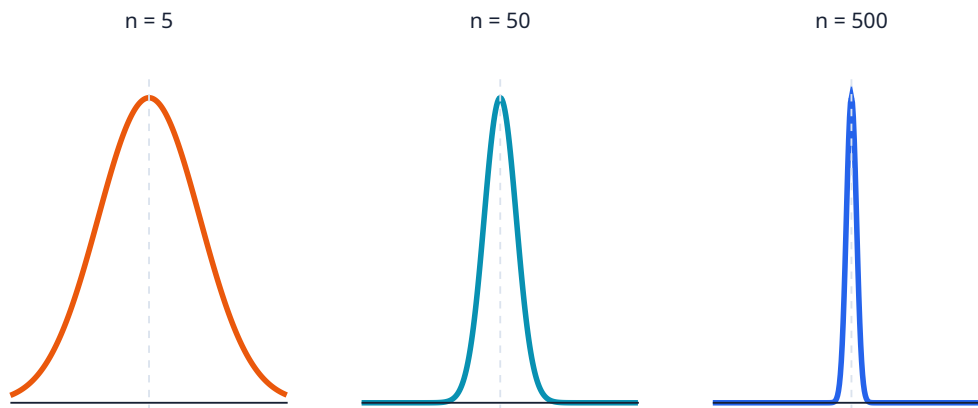
## 2. Simple random sampling, the picture (p003)



■ = the sampled items are spread randomly, NOT clustered  
 "without replacement" = once picked, an item is not put back

## 3. Standard error (p004)

### Sampling distributions narrow as n grows



centre stays at  $\mu$ ; widths use the exact  $1 / \sqrt{n}$  ratio

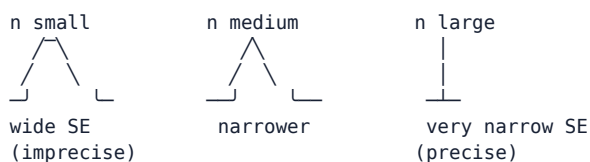
Sampling distributions narrow at the exact square-root rate

STANDARD ERROR measures the variability of a STATISTIC (usually the sample mean) across multiple hypothetical samples. It quantifies how much the estimate deviates from the true parameter.

$$SE = \sigma / \sqrt{n}$$

KEY: as  $n$  increases, SE DECREASES.  
Larger samples  $\rightarrow$  more precise estimates.

the shrinking spread:



## 4. WORKED: lightbulbs (p005)

$\sigma = 100$  hours.

Case 1:  $n = 25$      $SE = 100/\sqrt{25} = 100/5 = 20$  hours

Case 2:  $n = 100$      $SE = 100/\sqrt{100} = 100/10 = 10$  hours

THE SQRT RULE, memorise it:  
QUADRUPLING  $n$  HALVES the SE  
25  $\rightarrow$  100 is  $\times 4$ , so SE goes 20  $\rightarrow$  10 (halved)  
this is the answer to every "how large a sample" question

## 5. The Central Limit Theorem (p007-p010)

"If you take sufficiently large samples from a population, the distribution of the SAMPLE MEANS will follow a normal distribution, REGARDLESS of the population's underlying

shape (even if heavily skewed, uniform, or bimodal)."

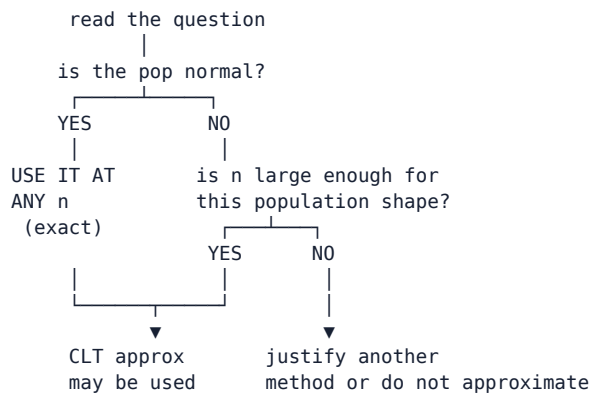
THREE RULES:

1. CENTRE      mean of the sample means =  $\mu$       ( $\mu_{\bar{X}} = \mu$ )
2. SPREAD     sd of the sample means =  $\sigma/\sqrt{n}$       ( $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ )
3. SHAPE      as  $n$  increases the shape becomes normal;  
                  $n \geq 30$  is the course rule of thumb

## 6. THE THREE CASES - the applicability conditions (p008)

1. NON-NORMAL population, large  $n$   
     -> sample means may be approximately Normal( $\mu$ ,  $\sigma/\sqrt{n}$ )  
      $n \geq 30$  is the course heuristic, not a theorem cutoff
2.  $n \leq 30$  AND the population IS normal  
     -> sample means are EXACTLY Normal( $\mu$ ,  $\sigma/\sqrt{n}$ )
3. Small  $n$  AND the population is NOT normal  
     -> do not assume a normal approximation without evidence

the decision tree:



## 7. The visual proof (p010a-p010b)

the 3x4 grid the slide shows:

|          | NORMAL                                                                              | UNIFORM                                                                             | SKEWED                                                                              | RANDOM                                                                              |
|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| POPULN   |  |  |  |  |
| $n = 10$ |  |  |  |  |
| $n = 30$ |  |  |  |  |

^ all four columns look NARROW BELLS at  $n=30$   
 that is the CLT: shape washed out, spread shrunk

## 8. WORKED: ATM wait times (p011-p012)

heavily right-skewed wait,  $\mu = 4$  min,  $\sigma = 2$  min,  $n = 36$ .  
 $P(\text{average wait} > 4.5 \text{ min})?$

STEP 1  $n=36 \geq 30$ , so the sample mean is normal (even though waits are skewed)

STEP 2  $SE = 2/\sqrt{36} = 2/6 = 0.333$

STEP 3  $Z = (\bar{X} - \mu)/SE = (4.5 - 4)/0.333 = 0.5/0.333 = 1.50$

STEP 4 upper tail at  $z=1.50$  -> 6.68%

the picture:



## 9. WORKED: impurity (p013-p014) - the ERRATA one

$\mu = 4.0 \text{ g}$ ,  $\sigma = 1.5 \text{ g}$ ,  $n = 50$ .  
 $P(3.5 < \bar{X} < 3.8)$ ?

$$SE = 1.5/\sqrt{50} = 0.212132$$

$$Z1 = (3.5 - 4)/0.212132 = -2.357 \rightarrow -2.36$$

$$Z2 = (3.8 - 4)/0.212132 = -0.943 \rightarrow -0.94$$

$$P = P(0 < Z < 2.36) - P(0 < Z < 0.94) \\ = 0.4909 - 0.3264 = 0.1645$$

ERRATA 6: the slide PRINTS  $Z2 = -0.4$ , which is WRONG.  
The correct value is  $-0.94$ .  
The slide's final  $0.1644$  is also wrong:  $0.4909 - 0.3264$  equals  $0.1645$  using rounded table areas. Using unrounded  $z$  values gives about  $0.1637$ . WRITE  $Z2 = -0.94$  and state which precision convention you used.

## 10. WORKED: LED bulbs (p015-p016)

"heavily skewed",  $\mu = 50000$ ,  $\sigma = 8000$ ,  $n = 64$ .  
 $P(\bar{X} < 48000)$ ?

$$SE = 8000/\sqrt{64} = 8000/8 = 1000$$

$$Z = (48000 - 50000)/1000 = -2.00$$

$$P = 0.0228$$

note the SAME  $-2.00 \rightarrow 0.0228$  as the normal example.  
It is the same table read, just reached through SE instead of  $\sigma$ .

## 11. WORKED: machine life, small n (p017-p018)

"the lives follow the NORMAL distribution",  $\mu = 7$ ,  $\sigma = 1$ ,  $n = 9$ .  
 $P(6.4 < \bar{X} < 7.2)$ ?

$$SE = 1/\sqrt{9} = 1/3$$

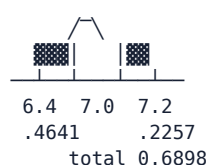
$$Z1 = (6.4 - 7)/(1/3) = -1.80$$

$$Z2 = (7.2 - 7)/(1/3) = 0.60$$

$$P = 0.4641 + 0.2257 = 0.6898$$

$n = 9$  IS BELOW 30, BUT THE QUESTION SAYS THE POPULATION IS NORMAL. That is case 2: the sample mean is EXACTLY normal. So we can still use the table. The word "normal" in the question gives an exact result, so no  $n \geq 30$  heuristic is needed.

the two tails added:



## THE SAMPLING/CLT KIT

|                       |                                      |
|-----------------------|--------------------------------------|
| SE of a mean          | $\sigma/\sqrt{n}$ (or $s/\sqrt{n}$ ) |
| SE of a proportion    | $\sqrt{p(1-p)/n}$                    |
| Z for a mean          | $(\bar{X} - \mu)/(\sigma/\sqrt{n})$  |
| Z for a single obs    | $(X - \mu)/\sigma$ <- NOT the same!  |
| $n \geq 30$           | course heuristic for CLT approx      |
| $n < 30$ , normal pop | exact normal                         |
| $n < 30$ , non-normal | normal approx needs justification    |

THE SINGLE COMMONEST MISTAKE IN THIS TOPIC:  
using  $\sigma$  in the denominator for a question about the AVERAGE  
of  $n$  items. An average ALWAYS divides by  $\sigma/\sqrt{n}$ .  
one bulb's life -> divide by  $\sigma$   
the MEAN of 25 bulbs -> divide by  $\sigma/\sqrt{25} = \sigma/5$